DESIGN BEFORE SCHEMA

Case Desk Flow

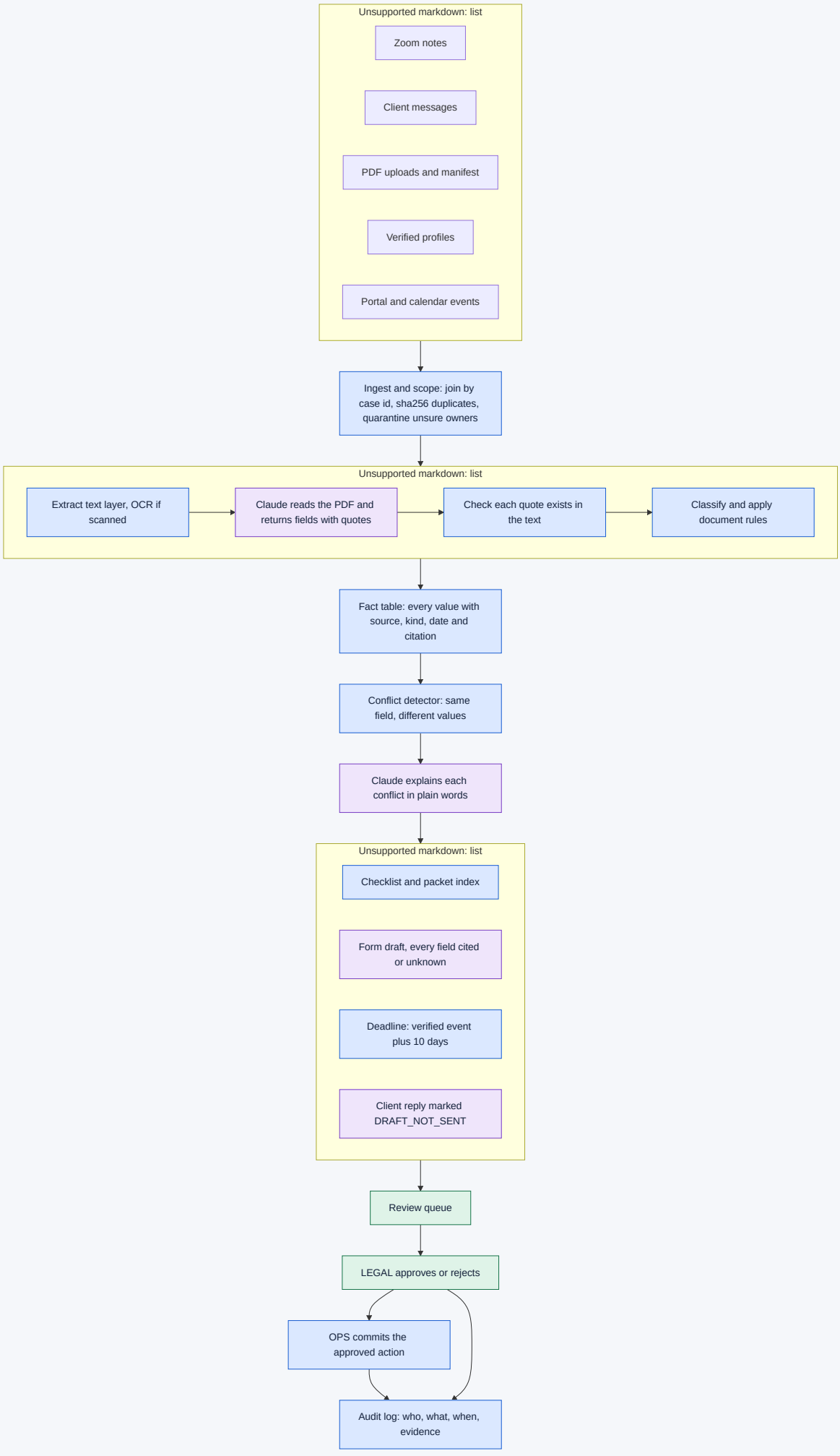
The whole path from raw intake to an approved, audited action, drawn before any table or route exists. Read figures 1 to 5 in order. The decisions at the bottom are what the schema will be built from.

Deterministic code Claude (language model) Human decision

FIGURE 1 · SYSTEM

Everything flows down one pipeline, and

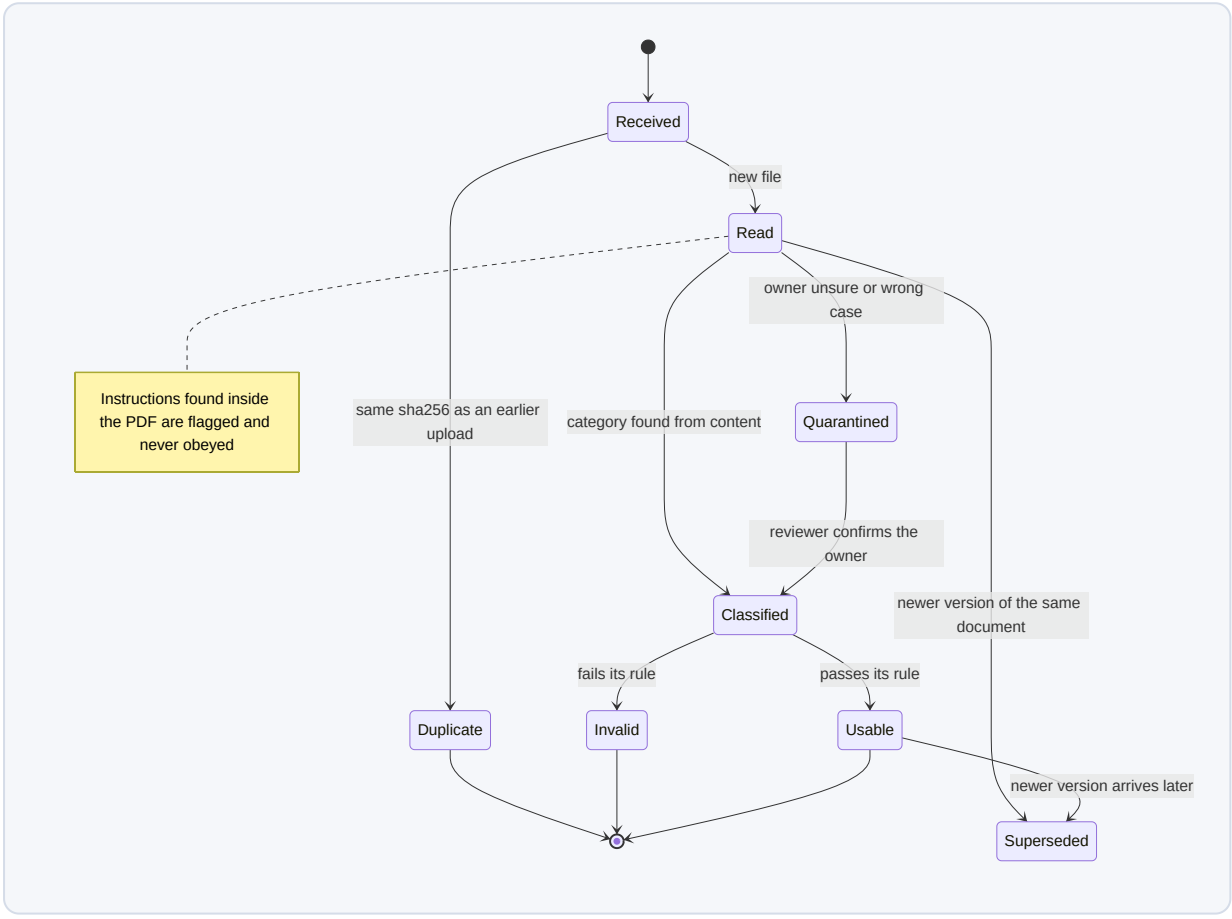
nothing is applied until a human approves


Unsupported markdown: list
Zoom notes
Client messages
PDF uploads and manifest
Verified profiles
Portal and calendar events
Ingest and scope: join by case id, sha256 duplicates, quarantine unsure owners
Unsupported markdown: list
Extract text layer, OCR if scanned
Claude reads the PDF and returns fields with quotes
Check each quote exists in the text
Classify and apply document rules
Fact table: every value with source, kind, date and citation
Conflict detector: same field, different values
Claude explains each conflict in plain words
Unsupported markdown: list
Checklist and packet index
Form draft, every field cited or unknown
Deadline: verified event plus 10 days
Client reply marked DRAFT_NOT_SENT
Review queue
LEGAL approves or rejects
OPS commits the approved action
Audit log: who, what, when, evidence

Claude appears in three places: reading a PDF, explaining a conflict, and drafting words. Every other step that could change a date, an identity or a permission is plain code, and the only way out of the proposal stage is through a LEGAL decision.

FIGURE 2 · ONE UPLOAD

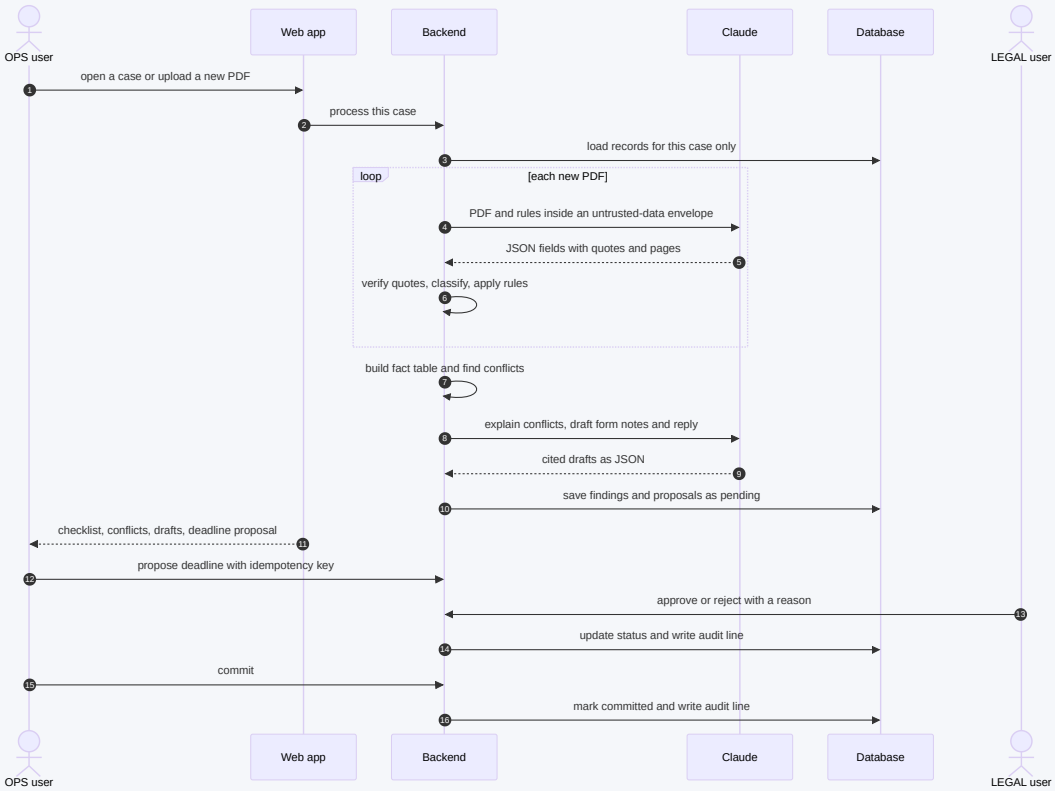
Uploaded is not usable: each PDF ends in exactly one outcome



The outcome and its reason are stored per upload, so the exception queue can show "UP-000007, duplicate of UP-000001" without recomputing. A judge's new PDF takes the same path as the 603 practice files.

FIGURE 3 · ONE RUN

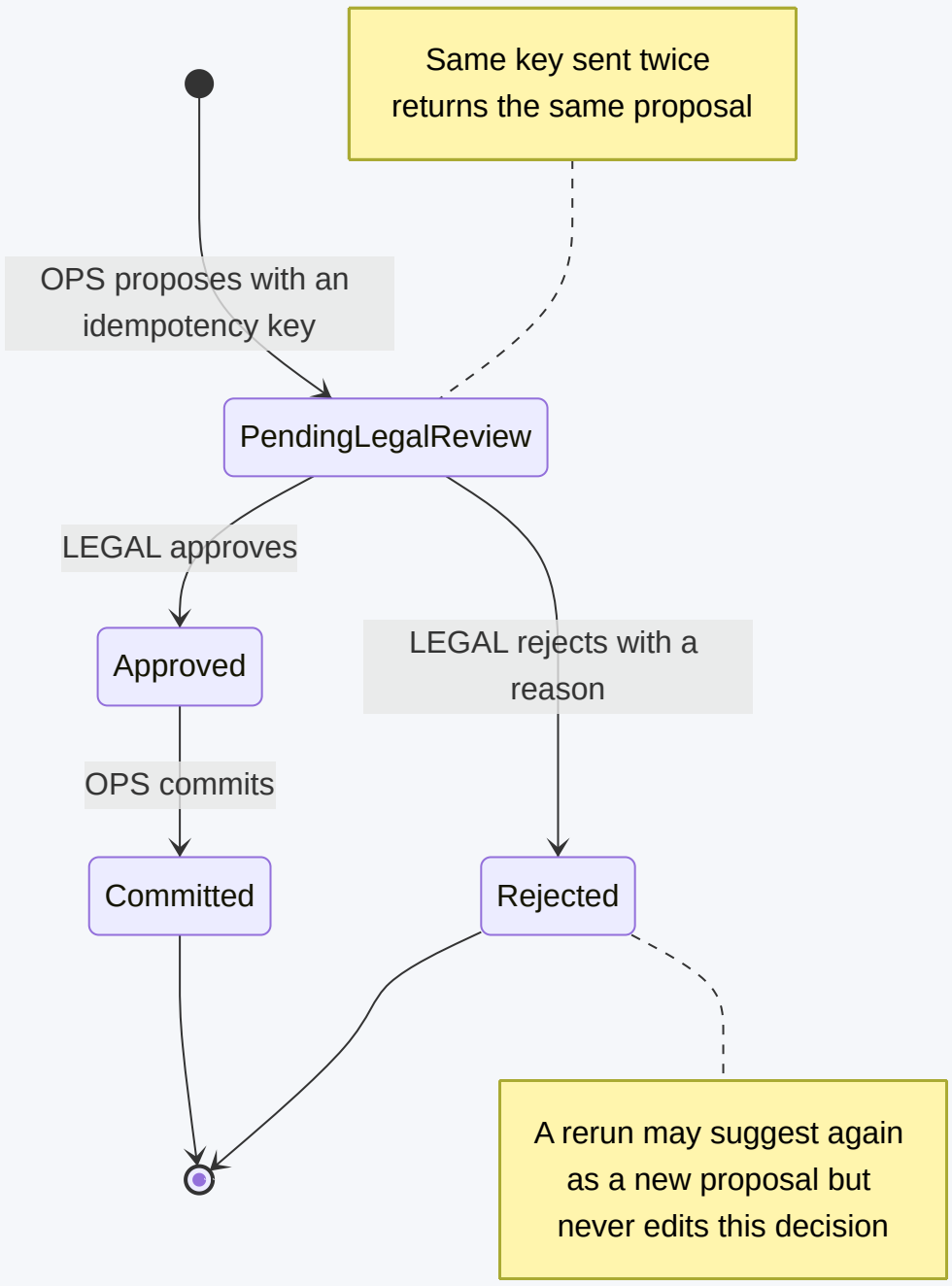
Who talks to whom when a case is opened and a deadline is approved



The model never touches the database. It returns JSON to the backend, which checks it and decides what to store. Step 4 is where case scoping is enforced, so no other case's data can reach the model.

FIGURE 4 · ONE PROPOSAL

The approval gate: states a proposal can be in



The same machine serves deadline activation, reviewed form export, packet approval and approved message status. Commit is refused unless the state is Approved, and the audit log gets one line per transition.

FIGURE 5 · SCREENS

What the user sees, in the order of the demo

Six screens, one actor switcher in the header. Wireframes show content, not styling.

1 Case list — /cases

Search case, client, receipt

Stage · Priority

Needs attention 12

CASE-0009 · INTAKE_REVIEW · OPS-05 · deadline in 1 day

CASE-0001 · INTAKE_REVIEW · 1 conflict

2 Case detail — /cases/:id

RESTRICTED · OPS-02 · LEGAL-02

Proposed deadline 2026-10-24 · calendar says 10-27

Sources · Documents · Conflicts · Checklist · Form · Timeline · Drafts

3 Documents — exception queue

UP-000001 · PASSPORT · USABLE

UP-000007 · PASSPORT · DUPLICATE of 01

UP-0000xx · letter · QUARANTINED

Client said "UNSPECIFIED" → read as EMPLOYER_LETTER

4 Conflicts — side by side

Surname CONFLICTED

Dawan · Zoom Z-0001 · claim

Dhawan · profile + UP-000001 · evidence

Open review request

5 Review queue — LEGAL view

ACT-000001 · DEADLINE_ACTIVATE · 2026-10-24

Trigger EV-0009-RFE (verified) + 10 days · rule SYN-DEADLINE-V1

Approve · Reject (reason required)

6 Audit trail — per case

PROPOSE · OPS-01 · 07:38

REVIEW_APPROVE · LEGAL-01 · 07:38

MOCK_COMMIT · OPS-01 · 07:38

Every line links to its evidence

TRUST ZONES

Who decides each thing

DECISION	CLAUDE	CODE	HUMAN
What a PDF says	reads, quotes	verifies the quote	
Which category it is, and is it usable	suggests	applies the rules	
Whose document it is		decides, quarantines if unsure	confirms
Do two sources conflict		compares values	
Why they conflict, in words	explains		resolves
Form field values	suggests with citations	checks citations	LEGAL appr
Deadline date		trigger + 10 days	LEGAL acti
Client reply	drafts	marks not sent	LEGAL appr
Who may do what		enforces	

BEFORE THE SCHEMA

Decisions to settle together

Each answer fixes a table or a route. My recommendation is first.

1. **Database.** SQLite for speed of setup; Postgres in Docker only if you expect parallel writers. One file is easier to reset between demo runs.
2. **One fact table or many.** A single `facts` table (case, field, value, source kind, source id, quote, date) makes conflict detection one query and keeps every citation uniform.
3. **Sync or background processing.** Run a case in a background task and let the UI poll for status. A judge's upload then shows progress instead of freezing.
4. **Where case scoping lives.** Every query takes a case id in one data-access layer, so no route can read across cases by mistake.
5. **Seeded review decisions.** Load `17_review_decisions_seed.jsonl` as real rejected proposals, so the CASE-0012 rerun test is just another case.
6. **Cache key.** Claude results keyed by file sha256 plus prompt version, so reruns are free and a prompt change invalidates cleanly.

> Next step once these are agreed: derive the tables from figures 2 and 4 (upload outcome, proposal state) and the fact table, then the routes from figure 3.